

ADVANCED QUANTUM MECHANICS

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Time dynamics

We will look at the time dynamics of the Schrödinger equation for a simple system exhibiting only two quantum states. The whole Hilbert space is then spanned by the orthonormal basis $\mathcal{B} = \{|0\rangle, |1\rangle\}$ such that unity is $1_2 = |0\rangle\langle 0| + |1\rangle\langle 1|$. Note that we do not specify what the wavefunctions $\langle x|0\rangle = \psi_0(x)$ and $\langle x|1\rangle = \psi_1(x)$ are as we focus only on the time evolution assuming these two functions are known from previously solving the stationary part.

Time dynamics of a driven two-level system

The system is described by the following Hamiltonian

$$\hat{H}(t) = -\frac{\hbar\omega}{2}\sigma_z + \hbar\Omega(\sigma e^{i\omega t} + \sigma^\dagger e^{-i\omega t}),$$

where ω characterizes both the level splitting and the drive frequency (resonant driving), and Ω is the coupling strength. The Pauli and ladder operators in the basis $\mathcal{B}$ are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and

$$\sigma = |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \sigma^\dagger = |1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

To get these matrix forms, we encoded the states as follows:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Check that applying σ to $|1\rangle$ leads to $\sigma|1\rangle = |0\rangle$ and $\sigma^\dagger|0\rangle = |1\rangle$. This places the $|0\rangle$ state at energy $-\hbar\omega/2$ and the $|1\rangle$ state at energy $+\hbar\omega/2$.

(i) Write the Hamiltonian $\hat{H}(t)$ in matrix form in the basis $\mathcal{B} = \{|0\rangle, |1\rangle\}$

(ii) Consider the general state

$$|\psi(t)\rangle = c_0(t)|0\rangle + c_1(t)|1\rangle.$$

Use the Schrödinger equation to find the coupled equations:

$$\begin{cases} i\dot{c}_0(t) = -\frac{\omega}{2}c_0(t) + \Omega e^{i\omega t}c_1(t), \\ i\dot{c}_1(t) = +\frac{\omega}{2}c_1(t) + \Omega e^{-i\omega t}c_0(t). \end{cases}$$

- (iii) Transform these equations into time independent equations by making the correct change of variables. Solve the equations of motion for the resonant case and initial condition $|\psi(0)\rangle = |0\rangle$. Show that the excited-state population evolves as

$$P_1(t) = |\langle 1|\psi(t)\rangle|^2 = \sin^2(\Omega t),$$

corresponding to *Rabi oscillations*.

Using the interaction picture

Consider the system above and now proceed differently:

- (i) Interaction picture that removes time dependence.

Split $\hat{H}(t) = \hat{H}_0 + \hat{V}(t)$ with

$$\hat{H}_0 = -\frac{\hbar\omega}{2} \sigma_z, \quad \hat{V}(t) = \hbar\Omega(\sigma e^{i\omega t} + \sigma^\dagger e^{-i\omega t}).$$

Define the carrier evolution

$$U_0(t) = e^{-\frac{i}{\hbar} \hat{H}_0 t} = e^{+i(\omega t/2) \sigma_z}.$$

The interaction-picture state and Hamiltonian are

$$|\psi_I(t)\rangle = U_0^\dagger(t) |\psi(t)\rangle, \quad \hat{H}_I(t) = U_0^\dagger(t) \hat{V}(t) U_0(t).$$

- (ii) Prove that $U_0^\dagger \sigma U_0 = \sigma e^{+i\omega t}$ and $U_0^\dagger \sigma^\dagger U_0 = \sigma^\dagger e^{-i\omega t}$ so that

$$\hat{H}_I(t) = \hbar\Omega(\sigma + \sigma^\dagger) = \hbar\Omega \sigma_x,$$

i.e. time independent in this picture.

- (iii) Closed-form solution and back-transform.

The interaction-picture evolution is generated by the time-independent operator $\hat{H}_I$. Show that:

$$|\psi_I(t)\rangle = e^{-\frac{i}{\hbar} \hat{H}_I t} |\psi(0)\rangle = e^{-i\Omega t \sigma_x} |\psi(0)\rangle = \left[\cos(\Omega t) \mathbb{1} - i \sin(\Omega t) \sigma_x \right] |\psi(0)\rangle.$$

Transform back to the Schrödinger picture:

$$|\psi(t)\rangle = U_0(t) |\psi_I(t)\rangle = e^{-i(\omega t/2) \sigma_z} \left[\cos(\Omega t) \mathbb{1} - i \sin(\Omega t) \sigma_x \right] |\psi(0)\rangle$$

Compare the solution with what you previously derived.